Descriptive Statistics - Measures of Central Tendency and variability

February 20, 2026

[1]:
```
import pandas as pd
```

[5]:
```
df=pd.read_csv('StudentsPerformance.csv')
```

[6]:
```
df
```

[6]:

	roll	name	class	marks	age	income
0	1	swapnali	TE	67.77	22.0	12000
1	2	sapna	TE	87.66	21.0	23000
2	3	manisha	BE	45.67	23.0	22000
3	4	sakshi	SE	76.66	NaN	13000
4	5	rohan	NaN	56.75	21.0	40000
5	6	ramya	BE	49.54	23.0	20000
6	7	soham	TE	NaN	21.0	12230
7	8	neha	BE	NaN	23.0	23111
8	9	arshad	BE	79.67	22.0	22000
9	10	manashri	TE	78.56	22.0	11000

[7]:
```
df.describe()
```

[7]:

	roll	marks	age	income
count	10.00000	8.000000	9.000000	10.000000
mean	5.50000	67.785000	22.000000	19834.100000
std	3.02765	15.468215	0.866025	8697.228121
min	1.00000	45.670000	21.000000	11000.000000
25%	3.25000	54.947500	21.000000	12422.500000
50%	5.50000	72.215000	22.000000	21000.000000
75%	7.75000	78.837500	23.000000	22750.000000
max	10.00000	87.660000	23.000000	40000.000000

[15]:
```
df.dtypes
```

[15]:

roll	int64
name	object
class	object
marks	float64
age	float64

```
income        int64
dtype: object
```

[18]:
```
df.mean(numeric_only=True)
```

[18] :
```
roll            5.500
marks          67.785
age            22.000
income      19834.100
dtype: float64
```

[19] :
```
df.median(numeric_only=True)
```

[19] :
```
roll            5.500
marks          72.215
age            22.000
income     21000.000
dtype: float64
```

[20] :
```
df.std(numeric_only=True) Traceback (most recent call last)
Cell In[46], line 1
----> 1 se=
```

[20] :
```
roll            3.027650
 marks         15.468215
 age            0.866025
income     8697.228121
dtype: float64
```

[21] :
```
df.min(numeric_only=True)
```

[21] :
```
roll            1.00
marks          45.67
age            21.00
income     11000.00
dtype: float64
```

[22] :
```
df.max(numeric_only=True)
```

[22] :
```
roll           10.00
marks          87.66
age            23.00
income     40000.00
dtype: float64
```

[23] :
```
import numpy as np
```

[24] :
```
np.std(df['marks'])
```

[24]: 14.469190544049102

[25] :
```
grl=df.groupby('class')
```

[26] :
```
te=grl.get_group('TE')
```

[27] :
```
te.min()
```

[27] :
```
roll                    1
name            manashri
class                 TE
marks              67.77
age                 21.0
income             11000
dtype: object
```

[28] :
```
te.max()
```

[28] :
```
roll                   10
name            swapnali
class                 TE
marks              87.66
age                 22.0
income             23000
dtype: object
```

[29] :
```
gr2=df.groupby('age')
```

[30] :
```
gr2.groups
```

[30]: {21.0: [1, 4, 6], 22.0: [0, 8, 9], 23.0: [2, 5, 7]}

[31] :
```
tw=gr2.get_group(22.0)
```

[32] :
```
tw
```

[32]:

	roll	name	class	marks	age	income
0	1	swapnali	TE	67.77	22.0	12000
8	9	arshad	BE	79.67	22.0	22000
9	10	manashri	TE	78.56	22.0	11000

[51]:
```
df=pd.read_csv('Iris.csv')
df
df.describe()
```

[51]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

[52]:
```
gr=df.groupby('Species')
```

[55]:
```
se=gr.get_group('setosa')
ve=gr.get_group('versicolor')
vi=gr.get_group('virginica')
```

[56]:
```
se.shape
```

[56]: (50, 6)

[57]:
```
ve.shape
```

[57]: (50, 6)

[58]:
```
vi.shape
```

[58]: (50, 6)

[67]:
```
print('\nsenota\n',se.describe())
print('\nversicolor\n',ve.describe())
print('\nvirginica\n',vi.describe())
```

senota

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.00000	50.000000	50.000000	50.00000
mean	25.50000	5.00600	3.418000	1.464000	0.24400
std	14.57738	0.35249	0.381024	0.173511	0.10721
min	1.00000	4.30000	2.300000	1.000000	0.10000
25%	13.25000	4.80000	3.125000	1.400000	0.20000
50%	25.50000	5.00000	3.400000	1.500000	0.20000
75%	37.75000	5.20000	3.675000	1.575000	0.30000
max	50.00000	5.80000	4.400000	1.900000	0.60000

versicolor

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.000000	50.000000	50.000000	50.000000

mean	75.50000	5.936000	2.770000	4.260000	1.326000
std	14.57738	0.516171	0.313798	0.469911	0.197753
min	51.00000	4.900000	2.000000	3.000000	1.000000
25%	63.25000	5.600000	2.525000	4.000000	1.200000
50%	75.50000	5.900000	2.800000	4.350000	1.300000
75%	87.75000	6.300000	3.000000	4.600000	1.500000
max	100.00000	7.000000	3.400000	5.100000	1.800000

virginica

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.00000	50.000000	50.000000	50.00000
mean	125.50000	6.58800	2.974000	5.552000	2.02600
std	14.57738	0.63588	0.322497	0.551895	0.27465
min	101.00000	4.90000	2.200000	4.500000	1.40000
25%	113.25000	6.22500	2.800000	5.100000	1.80000
50%	125.50000	6.50000	3.000000	5.550000	2.00000
75%	137.75000	6.90000	3.175000	5.875000	2.30000
max	150.00000	7.90000	3.800000	6.900000	2.50000

[ ]: